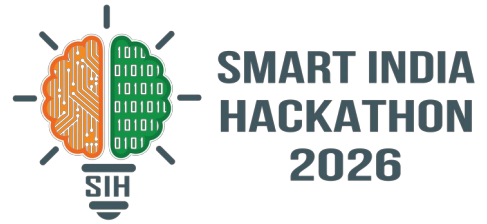


SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – 26175

**Problem Statement Title – Single-View
Height Estimation and Interactive 3D
Flythrough from Satellite Imagery**

Theme – Disaster Management

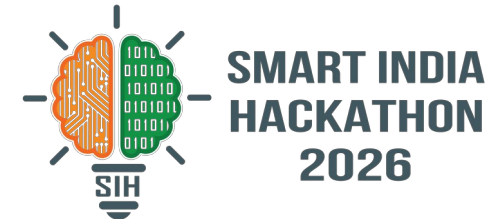
PS Category – Software

Team ID – [xxxxxxxxxxx]

**Team Name – [Team Name (Registered on
portal)]**



DepthWizard: One Satellite Image → Depth → Calibrated DSM → 3D Flythrough



Proposed Solution (Describe your Idea/Solution/Prototype)

- **DepthWizard**— an end-to-end pipeline that converts a SINGLE optical 2D satellite/aerial image into a relative depth map (rDSM), an estimated calibrated DSM (GeoTIFF) and an explorable 3D environment.
- Dual mode: non-georeferenced JPG/PNG → relative depth + 3D mesh; georeferenced GeoTIFF → CRS-preserving estimated DSM + height / slope products.
- Interactive 3D world: Inspect (orbit), Free-Fly drone mode (WASD + mouse) and automatic camera flythrough.

How it addresses the problem

- Removes dependency on costly stereo imagery, LiDAR sensors and InSAR — useful elevation geometry from ONE image, in seconds.
- Disaster response: reconstruct terrain and damage context when only a single post-event image exists.
- GIS-ready: GeoTIFF outputs preserve CRS / affine transform — drop directly into QGIS, ArcGIS, GDAL.

Innovation and uniqueness of the solution

- **First-of-its-kind** Fusion of Depth Anything V2 (relative geometry) + Sat3DGen (3D scene reconstruction) + GeoScale calibration layer ($DSM = DEM_{aligned} + \alpha \cdot D_{detail}$).
- Never silently converts relative depth to metres — metric values only after explicit calibration vs SRTM / GCPs (scientific honesty as a feature).
- Single unified open-source workflow: one image → survey-grade workflow → immersive browser flythrough.

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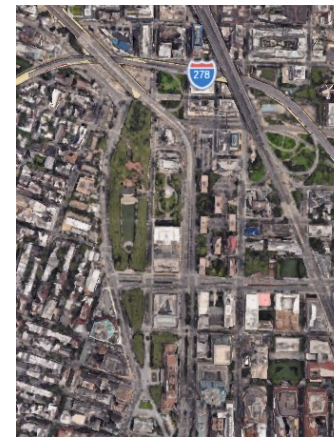
TECHNICAL APPROACH

Technologies to be used (e.g. programming languages, frameworks, hardware)

- Python 3.11 · PyTorch 2.x (AI inference)
- Depth Anything V2 — monocular relative depth
- Sat3DGen — satellite-image 3D reconstruction & mesh
- Rasterio / GDAL — GeoTIFF, CRS, affine transforms
- NumPy · OpenCV · SciPy — raster processing
- Three.js (WebGL) — browser 3D viewer
- FastAPI + React · Docker deployment

Methodology and process for implementation (Flow Charts/Images/ working prototype)

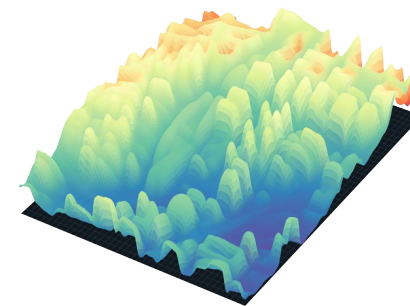
- ① Input handler — validates image, extracts CRS / bounds / transform
- ↓ ② Depth Anything V2 → relative depth (rDSM) | Sat3DGen → mesh + texture
- ↓ ③ GeoScale — align reference DEM (SRTM / Copernicus GLO-30), calibrate: $DSM = DEM + \alpha \cdot \text{detail}$
- ↓ ④ Analysis — height, slope, statistics → GeoTIFF / OBJ exports
- ↓ ⑤ Three.js viewer — inspect, free-fly, automatic flythrough



① INPUT
2D satellite image



② AI DEPTH
heatmap (rDSM)



③ OUTPUT
3D terrain + flythrough

**Working prototype: single-image → 3D terrain in < 2 s on consumer hardware
(live demo available)**

Analysis of the feasibility of the idea

- Working prototype already runs end-to-end: FastAPI backend + React/Three.js frontend, Dockerised one-click launch.
- 100% pretrained open-source models — no custom training data, no survey crews, offline-cached weights.
- Full synthesis pipeline completes in < 2 s on consumer hardware; automated pytest suite validates every endpoint.

Potential challenges and risks

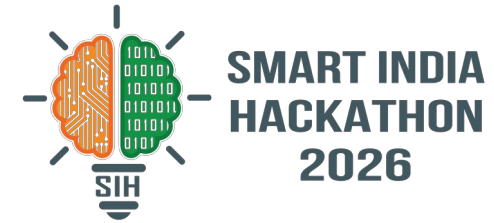
- Monocular AI depth is RELATIVE — not automatically survey-grade metric elevation.
- Coarse reference DEMs (SRTM 30 m) cannot validate individual building heights.
- Cloud cover, haze and occlusions degrade single-image inference; compute cost for very large scenes.

Strategies for overcoming these challenges

- GeoScale calibration: $DSM = DEM_aligned + \alpha \cdot D_detail$, or $Z = a \cdot D + b$ regressed from GCPs / EXIF telemetry.
- Explicit 'preview' vs 'calibrated' datum labelling — never silently convert relative depth to metres.
- Precomputed demo outputs guarantee resilience; validation via RMSE / MAE / correlation per land-cover class.

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IMPACT AND BENEFITS



Potential impact on the target audience

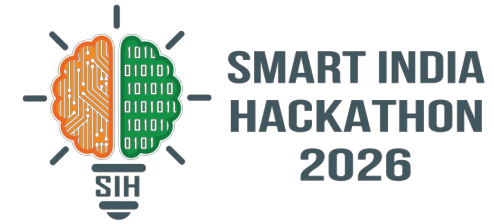
- ISRO / NDMA & disaster-response teams — terrain and damage context within seconds of a single post-event image, when no stereo pair or LiDAR pass exists.
- Urban planners, GIS & survey professionals — instant first-pass DSMs for zoning, line-of-sight and watershed analysis.
- Defence & infrastructure monitoring — rapid site intelligence in inaccessible or hostile terrain.

Benefits of the solution (social, economic, environmental, etc.)

- Social: faster flood, landslide and earthquake response planning from already-acquired archived imagery.
- Economic: eliminates cost of stereo flight campaigns, LiDAR sensors and multi-pass InSAR processing.
- Environmental: no additional sensor missions; enables continuous low-cost monitoring of fragile ecosystems.
- Strategic: open-source, GIS/CAD-interoperable outputs (GeoTIFF · OBJ · PNG · JSON) plug into national geospatial workflows.

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RESEARCH AND REFERENCES



Details / Links of the reference and research work

- Depth Anything V2 — L. Yang et al., NeurIPS 2024 — github.com/DepthAnything/Depth-Anything-V2
- Sat3DGen — satellite-image-based 3D scene reconstruction & mesh generation (project repository)
- MiDaS — R. Ranftl et al., “Towards Robust Monocular Depth Estimation”, ECCV 2020
- SRTM — T. G. Farr et al., “The Shuttle Radar Topography Mission”, Rev. Geophysics, 2007 · Copernicus GLO-30 DEM
- Rasterio — rasterio.readthedocs.io · GDAL — gdal.org · NumPy / SciPy / OpenCV
- Three.js — threejs.org/docs · FastAPI — fastapi.tiangolo.com · React — react.dev · Docker — docker.com
- SIH 2026 Problem Statement 26175 — ISRO · Theme: Disaster Management